

Homography Estimation on Ice Hockey Rinks

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Abstract

Transforming broadcast ice hockey footage into a normalized bird’s-eye view is a critical step for advanced sports analytics, enabling automated player tracking and tactical analysis. However, this task is complicated by the feature sparsity of the ice surface, frequent occlusions by players, and varying lighting conditions. This paper presents a robust, coarse-to-fine computer vision framework to estimate the homography between broadcast camera views and a standard 2D rink template. Our approach bypasses the limitations of purely local feature detection by employing a retrieval-based initialization strategy, where the input frame is matched to a pre-labeled reference database using Histogram of Oriented Gradients (HOG) descriptors. We subsequently isolate the structural “skeleton” of the rink using an adaptive feature extraction pipeline that suppresses crowd noise and enhances rink markings. Fine registration is achieved through a two-stage process: initial alignment via Masked SIFT matching followed by sub-pixel structural refinement using the Enhanced Correlation Coefficient (ECC) algorithm. Experimental results on NHL playoff footage demonstrate that our method achieves a mean reprojection error of 1.70 pixels, significantly outperforming general-purpose matchers like LightGlue in challenging scenarios involving heavy occlusion and symmetry ambiguities. This system provides a cost-effective, automated solution for generating accurate top-down reconstructions from standard broadcast feeds.

1. Introduction/Motivation

Ice hockey has evolved alongside viewing technology, transitioning from in-person spectating to broadcast television. Traditional training footage—whether from on-ice cameras or broadcast feeds, often suffers from occlusions and limited perspectives as players obstruct each other during play. Recent advances in computer vision now enable transformative bird’s-eye view reconstructions from broadcast im-

ages, overcoming these visual limitations.

1.1. What is the problem you are trying to solve?

The problem we are solving is to transform standard broadcast images of NHL ice hockey games into a normalized, top-down bird’s-eye view.

By detecting keypoints in the broadcast images such as rink markings, lines, and faceoff circles, matching them to corresponding locations in a reference top-down rink template, we can estimate the homography that maps image pixels onto the rink surface. This homography provides the transformation needed to project any camera view, despite changes in zoom or rotation, into a consistent geometric coordinate frame.

Several key challenges complicate the task of rink feature detection. First, the playing surface exhibits feature sparsity due to large, texture-less expanses of white ice, which provide limited visual information for keypoint detection. Additionally, occlusion by dynamic elements such as players, referees, and spectators frequently obscures the static geometric features of the rink. Finally, lighting variance including reflections and uneven illumination across the broadcast feed renders simple color-based filtering unreliable for consistent feature extraction.

1.2. Why is this interesting/important?

This problem is important because accurate homographies allow automated geometric reasoning on the game surface, including player tracking, puck localization, and tactical analysis, regardless of camera rotation or zoom. A reliable homography makes downstream sports analytics possible and greatly improves the robustness of computer-vision systems used in hockey broadcast and coaching applications. Assisting players to view their game at an angle never seen before helps train them for greater improvements in their spatial awareness on ice and more beneficial positioning against opposing teams.

2. Prior Work

2.1. What has been done in the prior literature to try to solve this problem?

To estimate the homography relating the camera plane to the playing surface, prior literature has primarily focused on three main approaches to feature extraction:

1. **Keypoint and Landmark Detection:** Traditional methods rely on extracting distinctive point features, such as corners and line intersections, using detectors like SIFT or ORB, followed by matching them to a template [1]. More recent approaches utilize deep learning object detection algorithms, such as YOLOv5, to identify specific semantic landmarks on the ice, such as face-off spots, goal posts, and line intersections [2]. To address the issue of sparse features in texture-less areas, some frameworks detect a grid of uniformly distributed keypoints across the entire field rather than relying solely on specific field markings [1].
2. **Semantic Segmentation and Line Detection:** Another prevalent method involves using semantic segmentation networks (e.g., VGG16, ResNet, or MANet) to classify pixels into categories such as field lines, face-off circles, and the ice surface [2][3]. Some approaches specifically target the extraction of lines and ellipses to match them with a synthetic database of edge images [4]. More advanced segmentation frameworks have recently proposed explicitly treating the "rink boundary" as a separate semantic class to improve geometric alignment [3].
3. **Direct Regression:** Instead of explicitly extracting intermediate features like points or lines, some deep learning approaches use Convolutional Neural Networks (CNNs), such as ResNet-18, to directly regress the homography parameters or the coordinates of four control points from the video frame [4]. These predictions are often refined using techniques like the Direct Linear Transform (DLT) or by minimizing alignment errors between the frame and the template.

2.2. What are current limitations of these methods?

Despite advancements, current methods face several significant limitations, particularly due to the specific visual nature of ice hockey rinks:

1. **Lack of Distinctive Features and Sparsity:** A major challenge is the uniform and texture-less appearance of the ice surface, which often lacks sufficient distinct features for accurate estimation [1]. Frames capturing open ice, such as the area around the blue lines, often lack high-fidelity keypoints, making accurate annotation and registration difficult [5]. Keypoint-based methods are particularly limited because sparse keypoints do not provide complete geometric information [3].
2. **Occlusion and Narrow Field of View:** In broadcast

footage, players frequently occlude critical visual cues like face-off spots or lines, leading to missing information in segmentation maps [1][3]. Furthermore, cameras often zoom in to follow the action, resulting in a narrow field of view that may contain fewer than the minimum four points required for homography calculation [1][2].

3. **Cost and Manual Effort:** Traditional manual analysis and hardware-based solutions (like installing fixed sensors) remain expensive and labor-intensive [2]. Even for creating training datasets, manual annotation of point correspondences is prone to human error and is time-consuming [5].
4. **Temporal Jitter and Boundary Neglect:** Methods that perform regression in a frame-by-frame fashion often suffer from projection jittering, where the estimated projection shakes over time due to a lack of temporal consistency [4]. Additionally, standard segmentation approaches often neglect rink boundaries because boundary pixels account for a very small fraction of the image; however, these boundaries are critical for spatial reasoning and are less likely to be occluded than internal markings [3].

2.3. Will your study potentially address any of these limitations? If so, how?

The study tackles the lack of distinctive features and sparsity by utilizing global descriptors rather than relying solely on sparse, local keypoints which often fail on texture-less ice. Specifically, it employs Histogram of Oriented Gradients (HOG) to match the input frame to a "nearest neighbor" reference frame, leveraging global shape structure (like the angle of the boards) instead of local texture. This is complemented by adaptive contrast enhancement (CLAHE) and adaptive thresholding, which enhance faint markings and ensure robust line detection even in feature-poor or unevenly lit areas. Furthermore, the issue of occlusion and narrow field of view is addressed through a retrieval-based initialization strategy. By inheriting a complete set of keypoints from a pre-labeled database match, the system can estimate the location of landmarks that are currently occluded or outside the frame. To handle spectators who often occlude the boards, an ROI Mask is implemented that mathematically fits a curve to the yellow kick-plate, effectively suppressing dynamic crowd noise while retaining the static rink structure. This robustness is confirmed by results showing lower reprojection errors compared to general matchers like LightGlue in scenes with heavy player occlusion. Regarding cost and manual effort, the solution is fully automated and designed for standard broadcast feeds, eliminating the need for expensive fixed camera systems or instrumented rinks. The automated pipeline handles feature extraction and matching without manual intervention, while the retrieval system bypasses the need for manual keypoint

identification on every frame. Finally, to mitigate boundary neglect and potential jitter, the study explicitly models the rink boundary by reconstructing the continuous edge from the yellow kick-plate and utilizes Robust Homography Estimation (RHO) with a strict inlier threshold to minimize geometric transfer errors.

3. Contributions

3.1. What specifically do you hope your study will contribute to the literature.

We hope our work contributes to the literature by advancing feature detection and perspective transformation methods for occluded, texture-sparse environments. The techniques developed here, adapted for sports broadcast imagery, provide a framework for generating tactical overhead views, addressing real-world challenges in sports analytics and broadening applications in computer vision. With the many challenges faced in field feature detections, we aim to show a robust method which avoids these problems and arrives at the solved solution.

3.2. What is the division of labour for your team. Explicitly identify the responsibilities of each team member.

- **Isakov:** Researched methods for transforming key points from broadcast images to an overhead rink perspective (Hough transform, Canny edge detection, MCMLSD). Manually removed the additional off-ice noise with fans within images. Responsible for the introduction and conclusion sections of the report and in ensuring the final project report was created in the correct format.
- **Habib:** Conducted comprehensive research on prior literature (Section 2) to identify current limitations in sports field registration, specifically regarding feature sparsity and occlusion. Additionally, performed empirical evaluation of OpenCV homography estimation algorithms (DLT, RANSAC, PROSAC) against the training dataset. Findings demonstrated the failure of standard SIFT on texture-less ice.
- **Matin:** Designed and implemented the first two stages of the pipeline: *Reference Frame Retrieval* and *Adaptive Feature Extraction*. This included deploying the HOG descriptor to identify global matches from the training database and engineering the crowd-suppression and adaptive segmentation logic to extract clean skeletons from raw broadcast frames.
- **Xingbang:** Designed the overall pipeline architecture, proposing a test-to-train retrieval strategy followed by a train-to-bird's-eye homography transformation. Initially explored rink yellow-line matching as the primary cue for reference frame retrieval and presented this idea in the project presentation. Additionally, conducted quanti-

tative evaluation using reprojection error and performed a comparative analysis between the proposed method and LightGlue-based image matching[6].

4. Dataset

For this study, we curated a dataset derived from broadcast feeds of official NHL games from the 2014 Stanley Cup Playoffs. The dataset consists of 31 frames in total, which were partitioned into two distinct subsets:

- **Training Set:** A set of 21 images used to build the retrieval database and learn the feature representations.
- **Test Set:** A set of 10 independent images used strictly for evaluation.

Each image in the dataset was manually annotated with pixel coordinates for distinctive key features, including the face-off circles, the center red line, and the goal lines.

5. Algorithm/System Description

The proposed system registers broadcast hockey footage to a bird's-eye template using a coarse-to-fine matching pipeline. The process begins by retrieving the most similar labeled frame from a training database (Coarse Initialization), isolating the rink structure (Feature Extraction), and refining the alignment using geometric constraints (Fine Registration).

5.1. Reference Frame Retrieval via Histogram of Oriented Gradients (HOG)

Identifying specific keypoints on a hockey rink is difficult because the markings are repetitive and symmetrical. Furthermore, even if a specific feature (e.g., a face-off circle) is successfully detected, calculating a homography based solely on that local region typically results in severe drift, where the alignment is accurate locally but degrades significantly across the rest of the rink. This issue is compounded by player occlusions and uneven arena lighting, which frequently hide critical landmarks.

To overcome these limitations without deep learning, we employ a retrieval-based strategy. Instead of relying on sparse, unreliable detections to build a transformation from scratch, we search our database for a training image that looks globally similar to the current broadcast frame.

Critically, each image in our database is paired with a set of ground-truth keypoints. By identifying the closest visual match, we effectively inherit these pre-calculated points as a reliable initial guess. This acts as a robust global reference, ensuring the alignment respects the entire field of view rather than over-fitting to a single visible feature.

- **Feature Representation (HOG):** We utilize the `cv2.HOGDescriptor` to extract global shape features. The Histogram of Oriented Gradients (HOG) descriptor operates by decomposing the image into small

spatial cells (e.g., 8×8 pixels). For each pixel within a cell, the algorithm calculates the gradient magnitude and orientation. These gradients are accumulated into a local 1-D histogram of edge directions. To account for illumination variations, local groups of cells are contrast-normalized over larger spatial blocks. By aggregating these normalized histograms into a single feature vector, HOG effectively encodes the global structure of the scene—such as the specific angle of the rink boards—while remaining invariant to local lighting changes and small deformations caused by player movement.

- **Dimensionality Reduction:** To ensure efficiency, both the training images and the incoming test frame are down-sampled to a fixed resolution of 64×128 pixels using `cv2.resize`.
- **Retrieval:** We compute the HOG feature vector for the test image using `hog.compute`. We then calculate the Euclidean norm (`numpy.linalg.norm`) between this vector and every pre-computed vector in our database. The training image with the minimum distance is selected as the Best Match, providing the initial keypoint labels. Figure 1 illustrates this retrieval process, showing the transition from the input test frame to the matched training image.

5.2. Adaptive Structural Feature Extraction

Once a reference frame is found, we must align it to the test frame. Because raw broadcast frames contain noise (crowd, ice texture), we employ a custom pipeline, `extract_features_adaptive`, to isolate the “skeleton” of the rink lines.

- **Preprocessing (Bilateral Filter):** We apply `cv2.bilateralFilter` rather than a standard Gaussian blur. This function smooths out high-frequency noise (like ice shavings) while preserving the sharp edges of the rink markings.
- **Contrast Enhancement (CLAHE):** To handle uneven arena lighting (shadows vs. spotlights), we convert the image to the LAB color space using `cv2.cvtColor` and `cv2.split`. We apply Contrast Limited Adaptive Histogram Equalization (`cv2.createCLAHE`) specifically to the L-channel. Unlike standard global equalization, CLAHE divides the image into small grid tiles and equalizes the histogram of each tile individually. This enhances local contrast in both dark and bright regions simultaneously.
- **Adaptive Segmentation:** We utilize `cv2.adaptiveThreshold` with the Gaussian method. Unlike global thresholding, this computes a unique threshold for each pixel based on its neighborhood (33×33 has been chosen in this project), ensuring robust line detection even in darker areas of the rink.

- **ROI Masking (Crowd Suppression):** To eliminate high-frequency noise from spectators, we implement `create_roi_mask`. We first detect the yellow kick-plate at the base of the boards using `cv2.inRange` in the HSV color space. Due to player occlusions, these detections are often fragmented. To reconstruct the continuous boundary, we extract the coordinates of the yellow pixels $\{(x_i, y_i)\}$ via `cv2.findNonZero` and fit a second-degree polynomial $P(x)$ using `numpy.polyfit`:

$$P(x) = ax^2 + bx + c \quad (1)$$

where coefficients a, b, c are minimized via least-squares. The final binary inclusion mask $M(x, y)$ is generated to strictly suppress the crowd while retaining the playing surface:

$$M(x, y) = \begin{cases} 255 \text{ (Valid)} & \text{if } y > P(x) \\ 0 \text{ (Ignored)} & \text{otherwise} \end{cases} \quad (2)$$

- **Structural Cleaning:** We refine the binary map using `cv2.connectedComponentsWithStats` to remove small noise blobs (area < 200 pixels) and apply `cv2.dilate` to thicken the remaining line segments for robust matching. The visual progression of this adaptive feature extraction pipeline is demonstrated in Figure 2.

5.3. Fine Registration Strategy

With clean structural skeletons extracted from both the test and training images, we perform a two-stage alignment process.

5.3.1. Stage 1: Masked SIFT Matching

We employ a “Masked SIFT” approach to find an initial set of correspondences.

- **Masking:** We combine the raw RGB image with our binary skeleton using `cv2.bitwise_and`. This ensures that feature detection is semantically constrained to the rink lines.
- **Detection & Description:** We initialize the detector with `cv2.SIFT_create` and extract keypoints using `sift.detectAndCompute`.
- **Matching:** Features are matched using `cv2.BFMatcher` (Brute Force) with the k-Nearest Neighbor method (`knnMatch`, $k = 2$). We filter outliers using Lowe’s Ratio Test.
- **Initial Estimate:** We compute a rough homography matrix (H_{init}) from these matches using `cv2.findHomography` with the `cv2.RANSAC` flag. This matrix serves purely as the coarse initialization required for the subsequent gradient-based refinement.

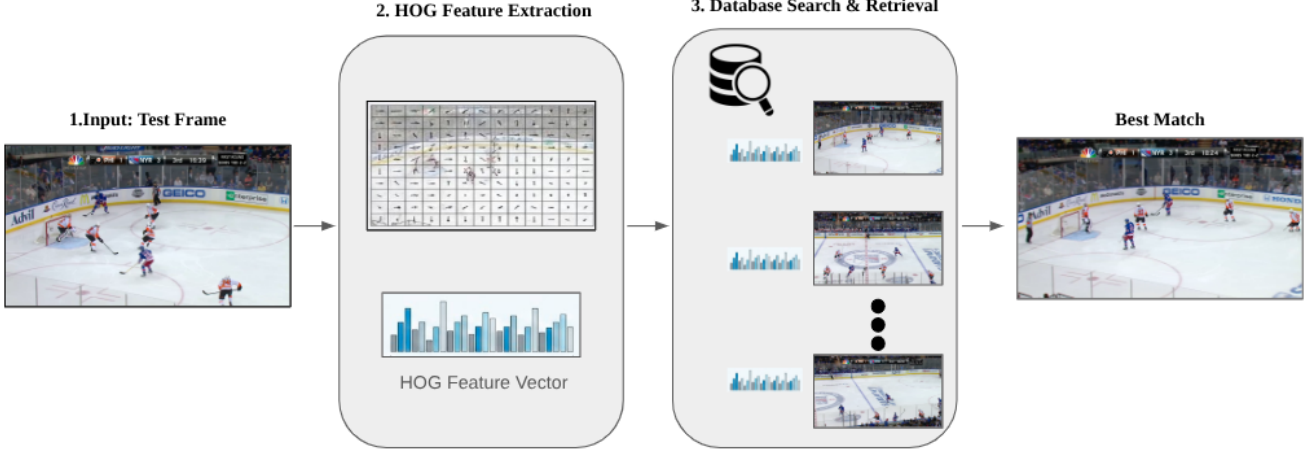


Figure 1. HOG-based retrieval process. The input test frame is processed to find the nearest neighbor in the training database to serve as a reference.

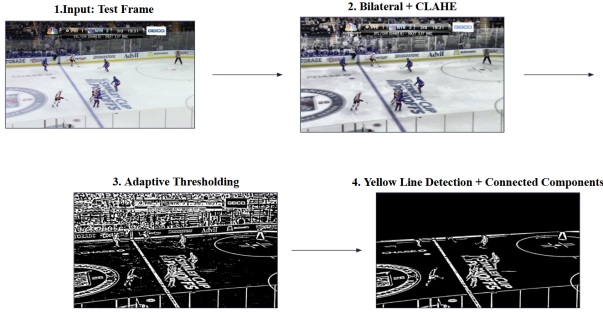


Figure 2. Adaptive Structural Feature Extraction pipeline. The sequence illustrates the transition from the raw input frame to the enhanced contrast image (CLAHE), the adaptive binary map, and finally the clean structural skeleton.

5.3.2. Stage 2: Structural Refinement (ECC)

SIFT matches can be sparse or slightly inaccurate. To achieve sub-pixel precision, we align the binary skeletons directly using the Enhanced Correlation Coefficient (ECC) algorithm.

- **Optimization:** We apply `cv2.GaussianBlur` to the skeletons to smooth the optimization landscape. We then utilize `cv2.findTransformECC`, initialized with the RANSAC homography (H_{init}). This iteratively perturbs the matrix to maximize the correlation between the training and test skeletons, yielding a highly accurate transformation matrix, $H_{refined}$.

5.4. Final Homography Estimation

The final homography mapping the broadcast view to the bird’s-eye template is computed by transferring the known training points to the test frame.

- **Point Transfer:** We utilize the refined ECC matrix to project the ground-truth keypoints from the training image ($\mathbf{p}_{train}$) onto the current test frame ($\mathbf{p}_{test}$). For the i -th keypoint in homogeneous coordinates:

$$\tilde{\mathbf{p}}_{test}^{(i)} = H_{refined} \cdot \tilde{\mathbf{p}}_{train}^{(i)} \quad (3)$$

This effectively labels the unlabelled broadcast frame with high-precision coordinates.

- **Robust Homography Estimation (RHO):** We compute the final homography (H_{final}) that maps these projected test points to the fixed bird’s-eye template points ($\mathbf{p}_{temp}$):

$$H_{final} = \text{findHomography}(\{\mathbf{p}_{test}\}, \{\mathbf{p}_{temp}\}) \quad (4)$$

We utilize `cv2.findHomography` with the `cv2.RHO` flag to minimize the reprojection error robustly. We employ a strict inlier threshold of 3.0 pixels. RHO is selected for its superior stability compared to standard RANSAC when handling geometric transfer errors.

6. Results

6.1. Qualitative and Quantitative Comparison

We quantitatively evaluate the accuracy of the estimated homographies using **reprojection error**, which measures the geometric discrepancy between key feature points projected by the estimated homography and those projected using the ground-truth transformation in the bird’s-eye coordinate system. This metric directly reflects the geometric alignment quality and is independent of appearance variations such as color, lighting, or player motion.

Table 1 shows the mean and standard deviation of reprojection errors for each test image. Most frames achieve low reprojection errors (below 3 pixels), indicating accurate

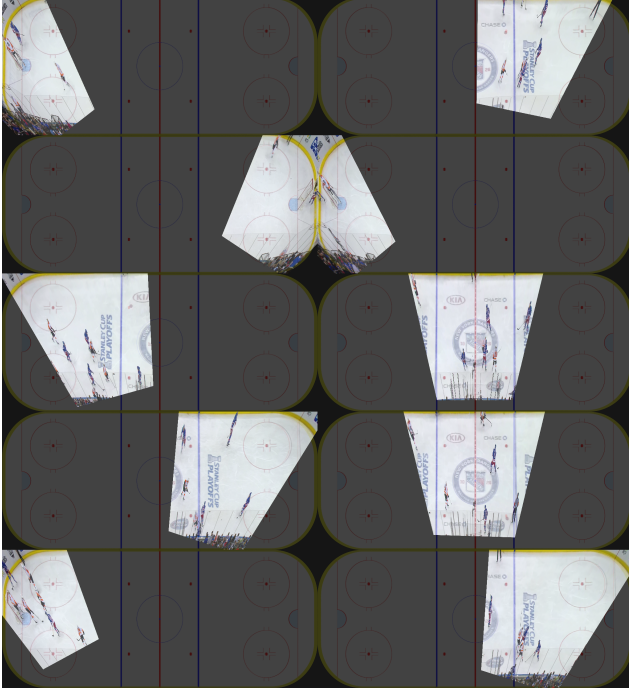


Figure 3. Qualitative Results: Broadcast Frames Warped to Bird's-Eye View

Table 1. Reprojection error statistics of our method (in pixels).

Image	Mean	Median	Std	Max
image1080.png	1.16	1.18	0.32	1.91
image1096.png	0.99	0.50	1.16	4.20
image1439.png	2.00	1.41	1.85	7.49
image241.png	1.65	1.52	0.54	2.69
image287.png	2.24	1.05	2.94	14.26
image293.png	1.43	1.02	0.91	3.63
image307.png	1.94	1.61	1.92	10.05
image350.png	2.73	2.50	2.31	8.63
image422.png	1.02	0.92	0.55	3.42
image466.png	2.25	2.08	1.37	5.83

and stable homography estimation, while a few challenging cases exhibit larger errors due to rink symmetry ambiguities. Across the test dataset, the reprojection error statistics are summarized as follows: the mean reprojection error is 1.70 pixels, the median error is 1.23 pixels, the standard deviation is 1.68 pixels, and the maximum error is 14.26 pixels.

The low median reprojection error (≈ 1.23 px) indicates that for the majority of test frames, the estimated homography aligns semantic rink landmarks very closely with the ground truth. The mean error (≈ 1.7 px) remains within an acceptable range for broadcast hockey footage, where rink

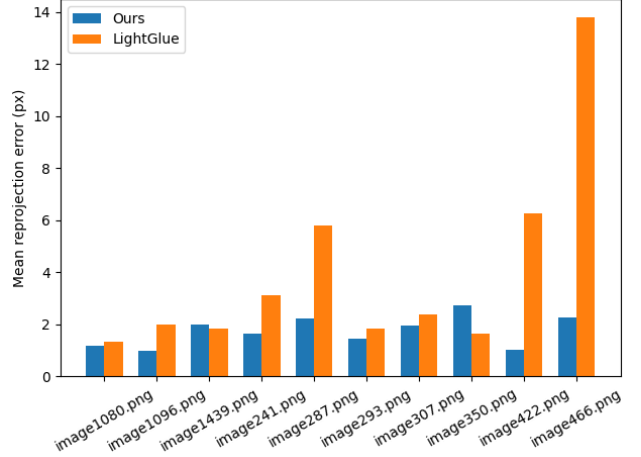


Figure 4. Per-image comparison of mean reprojection error between **Ours** and **LightGlue**. Each group of bars corresponds to one test image. Lower values indicate better geometric alignment.

lines and key markings typically span multiple pixels in the bird's-eye representation.

The relatively larger maximum error can be attributed to a small number of challenging cases, often involving extreme camera perspectives, partial rink visibility, or inaccuracies in manually annotated points near the image boundaries. Such outliers are expected under a single-plane homography assumption and do not indicate a systematic failure of the proposed method.

Overall, the reprojection error statistics demonstrate that the proposed system achieves stable and accurate geometric alignment across a wide range of test images.

Figure 4 compares the mean reprojection error between our method (**Ours**) and the LightGlue-based matching approach [6] across multiple test images. Overall, our method achieves consistently lower reprojection errors and demonstrates greater robustness across most test cases. In scenes with clear rink markings and limited occlusion, both methods perform comparably. However, in more challenging frames involving heavy player occlusions, partial line visibility, or rink symmetry ambiguities, the LightGlue-based approach exhibits significantly higher reprojection errors and larger variance.

These results indicate that our method is more robust to keypoint occlusion and structural ambiguity, making it better suited for homography estimation in real-world hockey broadcast scenarios. It is worth noting that LightGlue [6] is a general-purpose feature matcher and is not specifically designed for structured sports scenes with strong geometric symmetry.

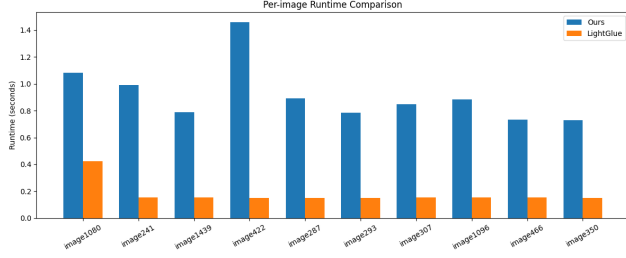


Figure 5. Per-image Runtime Comparison between Ours and LightGlue

6.2. Runtime Analysis and Efficiency Comparison

Figure 5 presents a runtime comparison between our method and the LightGlue-based matching approach. Overall, LightGlue achieves substantially lower runtime and exhibits highly consistent inference time across different test images. This is because of its neural network-based design, as discussed in the original LightGlue paper [6]. Specifically, LightGlue incorporates an adaptive inference mechanism that allows the network to early-exit when image pairs are intuitively easy to match, leading to nearly constant runtime for most test cases.

In contrast, our method used SIFT whose computational cost depends on image content and scene complexity. As a result, the runtime of our approach shows moderate variation across test images, reflecting differences in visible rink structures, feature distributions, and geometric configurations. Despite this variation, the runtime of our method remains within a practical range for offline evaluation, while providing improved robustness and geometric accuracy in challenging scenarios.

7. Conclusion

7.1. What are the main things you discovered in this project?

This project demonstrated the value of integrating multiple computer vision techniques to achieve robust results. By combining unconventional approaches such as the yellow border method with established techniques like SIFT and binary skeletonization, we created a hybrid pipeline that outperformed the accuracy of individual methods found in LightGlue. This synthesis of complementary approaches enabled the successful generation of bird’s-eye views from broadcast footage, highlighting the importance of methodological integration in solving complex computer vision problems.

7.2. What would you do in future work to extend the project?

Applying this concept onto other sports would be a very entertaining concept. Having a birds-eye view of a field in basketball, soccer, football, or even baseball would have very strong implications in how we view the game. Whereas in ice hockey there are borders and walls keeping the playing field concise and limited, all of these others sports are more extended and are played around the boundaries as well. Due to this there are many more objects obscuring the views of key features on each sports respective fields.

By taking a different path in extending this project, by implementing object recognition and real video transmission of broadcasted feed into a birds eye-view angles, we could apply player and puck tracking for fans and spectators to follow along. Introducing a highlight around their figures and showcasing their name and numbers once they come on or off the ice would be beneficial for all when watching live. Ultimately, while technology continues to transform how we view and analyze hockey, the essence of the game endures, uniting spectators and players alike across every viewing medium.

8. AI Usage

AI tools were strategically employed throughout this project to enhance efficiency and quality. During the planning phase, they assisted in identifying feasible and methodologically sound approaches. In the development phase, large language models (mainly Google Gemini) were utilized for code debugging and optimization, resulting in streamlined implementation. Finally, for documentation, AI writing assistants (specifically DeepSeek) were used for grammar correction, vocabulary enhancement, and ensuring concise, non-repetitive sentence structures, thereby improving readability and maintaining reader engagement.

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